MIL-I-19165B (WEP)
1 DECEMBER 1959
SUPERSEDING
MIL-I-19165A (AER)
28 FEBRUARY 1958

MILITARY SPECIFICATION

INDICATOR GROUP; AN/APA-125A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide a general purpose airborne indicator, displaying Radar, AMTI, Sonobuoy, Beacon and IFF video, singly or in certain useful combinations of displays as required in either AEW or ASW operations.

1.2 Classification - The Indicator Group AN/APA-125A shall be of one type and shall consist of the following:

Items	Type Designation	Max. Allow. Size (Inches)			Max. Allow. Wt. (Lbs.)	Appl. Para.
Indicator, Azimuth and Range	IP-414A/APA-125	20	12 3/4	28	114	3.4.1
Control, Indicator	C-2138/APA-125	22 1/2	7	9 3/4	12.5	3.4.2
Power Supply	PP-2192/APA-125A	13	10 3/4	19 1/4	40.0	3.4.3
Mountings						
Mounting	MT-1823/APA-125				6.0	3.4.1.2
Mounting	MT-1824/APA-125				2.5	3.4.3.2

1.3 Associated Equipment - The following associated equipments may operate with but shall not be supplied as a part of the Indicator Group AN/APA-125A:

Radar Identification Set	AN/APX-7 or AN/APX-20
Sonobuoy Receiver	AN/ARR-26A
Ground Position Indicator	AN/ASA-13
Radar Set	AN/APS-80

FSC 5895

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases, Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft Characteristics of
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

Bureau of Aeronautics

SAR-300	Material Changes and Material Bulletins; Preparation of

DRAWINGS

Air Force - Navy Aeronautical

MS 25245	Connections - Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, and/or 115/200-Volt Three-Phase AC, or combinations

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2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

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3.2 Design and Construction

3.2.1 General - This equipment shall conform with all applicable requirements of Specification MIL-E-5400 for design, construction and workmanship unless otherwise specified herein.

3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room ambient 25°C (+15°C or -5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Voltage	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combinations of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment.

3.2.3.1 Altitude - The complete equipment shall not be required to operate in excess of 30,000 feet.

3.2.4 Total Weight - The total weight of the equipment excluding cables shall not exceed 175 pounds.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 50 hours, continuously or intermittently, without necessity for readjustment of any controls which are inaccessible during flight.

3.2.6 Operating Life

3.2.6.1 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Interconnecting Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cable shall not be supplied as part of the equipment.

3.2.9 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-E-7894 having limits as specified hereinafter. The power required shall not exceed the specified amounts.

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(1) AC Power (Three Phase)

(a) 115/200 Volt Constant Frequency Type I

Operating Voltage Limits	108 to 121 Volts
Operating Frequency Limits	380 to 420 cps
Reduced Performance Voltage Limits	102 to 108 and 121 to 124 volts
Power (3ø)	1500 V. A.

(2) DC Power 28 Volts System

Operating Voltage Limits	25 to 29 Volts
Reduced Performance Voltage Limits	20 to 25 and 29 to 31 Volts
Power	4 amps

3.2.10 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.11 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum, however, overload protection shall be provided in primary A. C. power circuit.

3.2.11.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.12 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.13 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400. Unless otherwise specified, interchangeability shall be measured against a model supplied by the procuring activity.

3.2.14 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.15 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3 Performance -

3.3.1 Warm-Up Time - The warm-up time shall be approximately one minute controlled by an internal time delay relay.

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3.3.2. Inputs to the Indicator -

3.3.2.1 Synchronization Trigger - A positive polarity pulse shall be supplied to the indicator, coincident within 0.5 usec with the radar modulator trigger to initiate the timing circuits within the indicator. The pulse shall have the following characteristics:

(1)	Duration	0.6 to 5 usec.
(2)	Rise Time	0.3 usec between 10% and 90% of the normal peak voltage
(3)	Amplitude	5 to 100 volts peak
(4)	Pulse Repetition rate	100 to 2000 pps.

The indicator shall have a minimum recovery time of 300 usec. The sweep range gate, (see 3.3.3.1) shall be automatically decreased in width when the range selector switch setting is increased beyond the maximum possible for a particular pulse repetition rate. In the type of display providing open center operation of the sweep, no pretrigger shall be required.

3.3.2.2. Azimuth Synchro Data -

3.3.2.2.1 Antenna Azimuth Data - Three phase synchro data supplied by a Bendix Synchro Transmitter Type AY-201-1 or an approved equivalent shall be required by the Bendix Control Transformer AY-201-5 or an approved equivalent of the sweep servo within the indicator. This data, as selected by a switch on the indicator panel, may be stabilized as follows:

(1) Heading Stabilized - The data shall be referenced to the heading of the aircraft.

(2) North Stabilized - The data shall be referenced to North.

(3) Ground Stabilized - The data shall be referenced to North.

3.3.2.2.2 Cursor Azimuth - The input of three phase synchro data, supplied to a connector on the rear panel of the indicator, shall be required by the cursor azimuth servos to control the cursor bearings. This input would normally be supplied by the cursor azimuth control on the indicator panel, which is terminated in the same connector, by jumpers placed within that connector. In some cases, it may be desirable to have this data switched interchangeably with like synchro data generated in other future associated equipment. The connector provides a means of interrupting this link by a switch within the associated equipment and exciting the control transformer of the servo by either the cursor azimuth control or an external synchro transmitter of the same type. In addition, 26V, 400 cycle excitation voltage for three external synchros shall be provided.

3.3.2.3 Video Inputs - The following separate video inputs shall be provided:

(1) IFF Video, Mode I
(2) IFF Video, Mode II
(3) IFF Video, Mode III
(4) Sonobuoy Video
(5) Beacon Video
(6) Heading Marker Video
(7) Radar Video
(8) AMTI Radar Video
(9) Range Strobe #2
(10) Range Strobe #3
(11) Spare (unassigned)
(12) OPN CTR Video

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3.3.2.3.1 Input Characteristics - The video inputs shall have the following characteristics:

(1) Impedance - All video inputs shall be terminated within the equipment by a minimum of 2500 ohms.

(2) Level - The level of the video inputs shall be between 1.0 volt peak and 6.0 volts peak, positive polarity.

(3) Bandwidth - The video amplifier shall be capable of passing pulses of from 0.5 to 1000 usec duration. The output waveform shall have a maximum rise time of 0.2 usec for all video input channels. The droop of the output waveform shall be less than 1.0 db for the radar and spare video input channels. All other channels shall have an input coupling time constant of 50 usec.

(4) Fast Time Constant - Facilities shall be provided to allow the option of the above bandwidth, or a reduced input coupling time-constant of approximately 2 usec.

(5) Linearity - The input-output amplitude response shall be linear within 25% when the input is between 5% and 150% of the rated voltage.

(6) Noise - Video noise and hum generated within the video amplifier shall be less than 1% of its normal output voltage.

3.3.3 Indicator Display - PPI

3.3.3.1 Sweep Ranges and Durations

Range Selector Position	Sweep Speed Per Tube Radius	* Duration Minimum
10 MI	9.4 nautical mi ± .2 mi	36 nautical miles
20 MI	20 nautical mi ± 1/2 mi	80 nautical miles
50 MI	50 nautical mi ± 2 mi	200 nautical miles
150 MI	150 nautical mi ± 6 mi	350 nautical miles
250 MI	250 nautical mi ± 10 mi	350 nautical miles

* The duration of the sweep shall be controlled by a range gate. The gate generator shall have a minimum off time of 300 usec. The length of the gate, and therefore the duration of the sweep, shall automatically decrease should the pulse repetition period approach within 300 usec of the gate length.

3.3.3.2 Electronic Time Shared Cursors - Two electronic time shared cursors shall be provided. Selection of the display of no cursors, either or both cursors shall be possible. The interlacing of the cursor or cursors shall be controlled from the synchronizing trigger. If no cursors are displayed, no count-down of the sweep shall be made. When one cursor is being displayed, it shall be interlaced 1:15 with the sweep. When two cursors are displayed they shall be interlaced 1:1:14 with the sweep. The origins of each cursor shall be intensified to assist in establishing initial positions. Also, provisions shall be made to reduce the interlace time to 1:7 and 1:1:6 respectively by the inclusion of a maintenance switch. On the 50, 150 and 250 mile sweep ranges, the writing speed of cursors #2 and #3 shall be twice that of cursor #1, or approximately 6.18 usec/mile. On the 10 and 20 mile ranges the writing speed of all cursors shall be the same, or 12.36 usec/mile.

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3.3.3.2.1 Cursor Azimuths - The azimuth of each cursor shall be controlled by a servo driven resolver as described in 3.4.1.3.2.1. Normally the cursor azimuths are controlled by individual bearing cranks with digital counters on the front panel. However, provision is made for remote or local azimuth control as described in 3.3.2.2.2.

3.3.3.2.2 Cursor Range Strobes - An electronic strobe, controllable in intensity, shall be generated within the indicator for each cursor. The strobe range shall normally be controlled by a multiturn precision potentiometer, adjusted by a range strobe handcrank and indicated by a digital counter on the front panel. The maximum range of the strobe shall be 697 miles on the 50, 150, and 250 mile ranges at a speed of 6.18 usec/mile, and the accuracy of the strobe shall be 1 mile or 1% of the range, whichever allows the larger error. The connection of the potentiometers to the strobe generators shall be made through a jumper in the connector and plug on the rear panel of the indicator. Thus, selection of local or remote operation is possible when this jumper is removed and replaced by a switch in an associated equipment. A similar potentiometer could be included in this associated equipment to control remotely the range of each strobe. Separate video inputs are provided to allow the mixing of externally generated strobes with internal strobe video. These strobes should be approximately 0.5 usec wide, 2 to 6 volts in amplitude, maximum rise time of 0.2 usec, obtained from a 100 ohms source. The maximum range of the strobe shall be consistent with the gate length and shall be at a speed of 12.36 usec/mile for the 10 and 20 mile ranges and 6.18 usec/mile for the 50, 150, and 250 mile ranges.

3.3.3.2.3 Cursor Origin Slew - The origin of each cursor may be at the origin of the sweep or at any point on the display as described below. When cursor slew operation is selected by placing the cursor slew switch in the IN position, the origin of each cursor shall be controllable by means of geared two speed, uncalibrated joysticks defined in 3.3.4. The amount of origin slew shall be as follows:

Range	Coarse Control	Cursor Slew Fine Control
250 MI	±2.0 radii	±0.2 radii
150 MI	±3.33 radii	±0.33 radii
50 MI	±10.0 radii	±1.0 radii
20 MI	±5.0 radii	±0.5 radii
10 MI	±10.0 radii	±1.0 radii

The displacement of the cursor origins from the sweep origin shall remain constant in miles as the range selector is changed between the 10 MI, and 20 MI range positions and also between the 50 MI, 150 MI, and 250 MI range positions.

3.3.3.2.4 Alternate Cursor Origin Slew - The cursor slew voltage inputs, normally obtained from the joysticks potentiometer in the Indicator Control, may be obtained from an associated equipment and supplied through the connectors on the Indicator Control. Under these conditions, the scale factors of the input voltages shall be the same as those described for the sweep slew. (see 3.3.3.4)

3.3.3.2.5 Registration - With the cursors superimposed upon the main PPI sweep, externally applied range markers shall coincide within 1.25% of the selected sweep range under conditions of para. 3.2.2 and within 2% of the selected sweep range under conditions of para. 3.2.3.

3.3.3.3 Combined Origin Slew - Provisions shall be made to allow the selection of combined slew to off-center the origins of the sweep and cursors together with one set of controls. On the 250, 150, and 50 mile ranges the maximum combined slew shall be a minimum of 300 miles. This allows the observation of any visible portion of these sweeps including 100 mile radius of GPI off-centering. On the 20 and 10 mile range positions the maximum combined slew shall be 65 miles minimum. The combined origin slew shall be controlled by a two-speed joystick, described in 3.3.4 mounted on the control panel. Maximum fine combined slew shall be a minimum of 6 1/2 miles on the 10 and 20 mile position and a minimum of 28 miles on the 50, 150, and 250 mile positions.

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3.3.3.4 Sweep Slew - Off-centering circuits shall be provided to accept two-wire inputs of NS displacement and EW displacement voltages from an associated GPI equipment. The origin of the sweep shall be off-centered when the stabilization selector is in the GROUND position in the direction indicated 1 mile for each 1 volt of GPI input up to ± 150 miles from a given reference or operation point. It shall be possible, in this associated equipment, to reset these inputs to any ground reference while in flight. The operating voltages for the potentiometers (plus and minus 150 V DC at 100 ma maximum) shall be supplied from the indicator power supply. The accuracy of displacement of the sweep origin with application of an external slew voltage shall be ± 2% of the selected sweep range.

3.3.4 Joystick Controls - Two speed geared joysticks shall be provided to control the off-centering circuits described above, allowing the simultaneous origin slew in the NS and EW directions proportional to the distance the handle is moved. A switch shall be incorporated so that the selection of the coarse-fine operation may be made on the joystick handle. With this switch not operated, the fine operation shall be selected. Operating the switch shall place the joystick in coarse operation. When a coarse adjustment of origin displacement has been made and the button released, the position shall be held, allowing a fine adjustment about this point.

3.3.5 IFF Controls - The indicator shall include controls on the panel, allowing the selection of one or more of three IFF modes of operation of the associated equipment. When the selection of a particular mode of operation is made, the control shall also cause the video input corresponding to that mode to be placed into the video mixer. The selection of mode of operation shall be accomplished by making a connection between a pin of a connector on the rear panel of the indicator corresponding to that mode and a common pin.

3.4 Detail Requirements

3.4.1 Azimuth and Range Indicator IP-414A/APA-125 - The indicator unit shall meet the following requirements:

3.4.1.1 Function - The function of the indicator is to provide a general purpose airborne display of Radar, AMTI, IFF, Sonobuoy, Beacon and OPN CTR Video singly or in certain useful combinations with provisions for retrace insertion. Two electronic time shared cursors are available for plotting selected data integral with the video display. The display may be stabilized in azimuth to North, heading or ground references and the origin of the sweep and cursors may be off-centered individually and simultaneously. The group equipment is primarily intended for installation in aircraft assigned AEW or ASW missions.

3.4.1.2 Mounting - The Mounting MT-1823/APA-125 shall support and isolate the indicator from shock and vibration, and shall meet all the applicable requirements of Specification MIL-C-172.

3.4.1.3 Contents of Unit - The unit shall contain the following circuits, controls, sub-assemblies.

3.4.1.3.1 High Voltage Supply - A high voltage supply shall be contained in the unit to supply the first and second anode potentials of the CRT. This assembly will operate with an input of single phase (115 V AC ± 10% 380-420 cps) from the power supply with a one minute delay. The assembly then provides 8 KV DC ± 1% regulated by an external shunt regulator, and 650 $^{+100}_{-50}$ V DC.

3.4.1.3.2 Servo Drive Assemblies - The unit shall contain three identical servo drive assemblies which provide azimuth resolution for the sweep and the two cursors. The assembly shall be electrically connected to the mounting by means of a subminiature plug and shall contain the following components:

(1) Servo Motor

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Table III

(2) Synchro Control Transformer, AY-201-5 or an approved equivalent.

(3) Sine-Cosine Resolver

(4) Required casting, bearings and gears.

3.4.1.3.2.1 Azimuth Servo and Sweep Resolvers - The azimuth synchro input described in 3.3.2.2 shall excite a control transformer, type AY-201-5 or equivalent. The servo system shall be capable of rotational speeds from zero to 30 rpm with a maximum velocity error of 30 minutes. It shall be capable of sector scanning at rates corresponding to 15 rpm.

3.4.1.3.3 Chassis Assemblies - Insofar as is practicable, the circuits for the generation of sweep voltages, strobes, time-sharing counters and gates, off-centering voltages, sweep driver amplifiers, servo amplifiers, and video mixers and amplifiers shall be contained on detachable, sub-chassis assemblies. Electrical connections shall be made through the use of acceptable connectors to facilitate the removal of the assembly. The maximum possible access to tubes and other components shall be provided consistent with space, size and weight limitations.

3.4.1.3.4 Indicator Tube - The indicator tube shall be a 10-inch, aluminized magnetic focus CRT with P-7 long persistence phosphor screen. It shall be provided with a reflection plotter mounted over the screen of the tube. This plotter shall be of plexiglas, colored corresponding to Rohm and Haas filter color designation #2208, or an approved equivalent. The plotter shall be edge lit, and shall be engraved with an azimuth scale and the center shall be marked with a cross-hair. A dimmer control shall provide for the adjustment of the intensity of illumination of the plotter. Provision shall be made for the connection to an associated photographic camera of electrical contacts representing the range selector position. These range position connections shall also be available on a connector on the rear panel.

3.4.1.3.5 Indicator Panel - The unit shall be supplied with an edge lighted panel in accordance with MIL-P-7788 except that all panel light filters shall be clear. The intensity of illumination shall be controllable by a dimmer control. This panel shall mount the necessary controls.

3.4.1.3.6 Front Panel Controls - The following controls, labeled as indicated, shall be available on the front panel:

(1) "CURSOR SELECTOR: OUT, 2, 3, 2-3 - This control shall permit the display of either or both cursors on the indicator tube.

(2) "POWER: ON, OFF" - A switch used to control the power to the indicator group.

(3) "MILES-STROBE RANGE 2" - This control shall provide adjustment of strobe #2 along the cursor and have provisions for indicating the range of the strobe.

(4) "RANGE: 10 MI, 20 MI, 50 MI, 150 MI, 250 MI" - This control shall provide selection of displayed range on the indicator.

(5) "○ PANEL - ● DIAL" - These controls shall allow for the adjustment of the panel illumination and dial illumination independent of each other.

(6) "MILES -STROBE RANGE 3" - This control shall be identical with (3) above but shall control strobe #3.

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(7) "CURSOR INT" - This control shall allow the adjustment of the intensity of the cursors and shall be compensated so that a change in sweep speed shall not require a readjustment of the intensity control.

(8) "SWEEP INT" - This control shall allow the adjustment of the intensity of the sweep and shall be compensated so that a change in sweep speed shall not require a readjustment of the intensity control.

(9) "FOCUS" - This control shall provide satisfactory adjustment of the focus of the sweep trace on the indicator tube.

(10) "MASTER GAIN" - This control shall allow the adjustment of the video level out of the mixer into the amplifier.

(11) "STROBE INT; ● INT 2, ▲ INT 3" - These controls shall allow the adjustment of the strobe video intensity displayed on the cursors. They shall allow the setting of the level of the strobe in each cursor independently.

(12) "DEGREES - CURSOR BEARING: 3" - This control shall allow the bearing of Cursor # 3 to be varied by 360° and shall have provisions for mechanically indicating the bearing.

(13) "DEGREES-CURSOR BEARING: 2" - This control shall be identical with (12) above and shall control cursor # 2.

(14) "SPARE: IN, OUT" - (Spare Video Control)

(15) "FTC: IN, OUT" -

(16) "RADAR, OUT, AMTI" -

(17) SONO, OUT, BEACON" -

(18) "OPN, CTR, OUT, HM" -

(19) "IFF-MODE I: IN, OUT" -

(20) "IFF-MODE II: IN, OUT" -

(21) "IFF-MODE III: IN, OUT" -

3.4.1.3.7 Video Level Adjustment - In addition to the above, video level adjustments shall be provided for video input balance control. The following adjustments, mounted on the front panel, shall have a hinged locked cover protecting them from tampering or accidental misalignment.

(1) "RADAR-AMTI"

(2) "SONOBUOY - BEACON"

(3) "OPN CTR"

(4) "HM"

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(5) "IFF"

(6) Spare

3.4.1.4 Maintenance Adjustments and Test Points - Maintenance adjustments shall be restricted to only those functions which must be made infrequently. These adjustments shall be made accessible to the maintenance personnel without requiring the opening of interlocks or exposure to dangerous voltages.

3.4.1.5 Accuracy of Presentation

3.4.1.5.1 Sweep Jitter - The sweep jitter observed at the end of the sweep displaced 4 radii from tube center shall not contribute to the sweep trace width an increase of more than 3/64 inch.

3.4.1.5.2 Origin Jitter - The jitter of the origin of any sweep shall not exceed 1/32 inch.

3.4.1.5.3 Origin Drift - The displacement of the origin shall not exceed 1/8 inch for all environmental conditions encountered for a period of at least 6 hours.

3.4.1.5.4 Linearity of 10 Mile Sweep, Centered Presentation - The linearity of the 10 mile sweep up to 800 yards range shall be the best possible, consistent with the state-of-the art. From 800 to 20,000 yards, the linearity at any azimuth position shall be ± 200 yards. Beyond 20,000 yards the linearity shall be ± 1% of the range.

3.4.1.5.5 20, 50, 150 or 250 Mile Sweep, Centered Presentation - The linearity of the sweep at any azimuth angle and using any of the other sweep ranges, shall be ± 1% of the range.

3.4.1.5.6 Circularity - Circularity shall be maintained so that a deviation from a true circle at one radius shall not be greater than ± 2% in range at any azimuth angle.

3.4.1.5.7 Azimuth Positioning - Centered Presentation - The indicated azimuth at any point shall be ± 2° of the azimuth input information.

3.4.1.6 Video Mixing - Provisions shall be made to mix any or all of the following inputs by panel selection switches described in 3.4.1.3.6 (14) to (21) inclusive. Mixing may be additive or non-additive.

(1) Radar or AMTI

(2) Sonobuoy or Beacon

(3) OPN CTR and Heading Marker or Heading Marker

(4) Spare Video

(5) IFF Video, Mode I

(6) IFF Video, Mode II

(7) IFF Video, Mode III

3.4.1.6.1 Remote Video Mixing - Provisions shall be made for control of mixing in a remote video mixer, of any or all of the inputs of 3.4.1.6, by panel selector switches.

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3.4.1.7 Indicator Display - Open Center Presentation - When the OPN CTR selector switch is operated, the PPI shall automatically open center and shall off-center by preset amounts. Provisions shall be made within the indicator to operate open centered without the requirement of pre-triggers from the radar system. The visible sector displayed in OPN CTR operation shall not be fixed in design. Maintenance adjustments shall be provided to allow variation of the magnitude of open centering from zero to 15 miles and of off-centering from zero to 20 miles. The cursors shall originate at the start of the sweep and shall be open centered. The display shall be heading stabilized and present heading marker video. All joystick controls shall be disabled during this mode of operation.

3.4.2 Control, Indicator C-2138/APA-125 - The indicator control shall meet the following requirements:

3.4.2.1 Function - This unit shall provide additional functional control of the indicator display.

3.4.2.2 Mounting - The unit is to be mounted on the lower part of the front panel of the indicator by means of a hinged bracket. The panel shall be capable of being hinged upward to allow clearance for the operator when he takes his position. The hinge mounting shall be easily detached from the indicator unit for maintenance purposes.

3.4.2.3 Controls - The unit shall mount the two cursor joystick controls and the combined sweep slew joystick control. The following switches and controls shall be mounted on the control panel, and labeled as indicated.

(1) "STABILIZE: GND, NORTH, HEADING"

(2) " CURSOR SLEW 2"

(3) "CURSOR SLEW 2: IN, OUT"

(4) "COMBINED SLEW"

(5) "COMBINED SLEW: IN, OUT

(6) "CURSOR SLEW 3"

(7) "CURSOR SLEW 3: IN, OUT"

3.4.2.4 Edge Lighted Panel - This unit shall be supplied with an edge lighted panel in accordance with Specification MIL-P-7788 except that all panel light filters shall be clear.

3.4.3 Power Supply PP-2192/APA-125A - The power supply shall meet the following requirements:

3.4.3.1 Function - The power supply shall provide AC and regulated DC power to the indicator unit.

3.4.3.2 Mounting MT-1824/APA-125 - This unit shall support and isolate the power supply from shock and vibration, and shall meet all the applicable requirements of Specification MIL-C-172.

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3.4.3.3 Output Voltages - The unit shall supply the following voltage outputs within rated regulation from 10% to 110% of rated load at given temperature within the environmental limits.

	VOLTAGE	VOLTAGE TOLERANCE	REGULATION	MAX. PK TO PK RIPPLE	RATED CURRENT
(1) Regulated	+150 V dc	± 1%	0.1%	30 mv	200 ma
(2) Regulated	-150 V dc	± 1%	0.1%	30 mv	400 ma
(3) Regulated	+250 V dc	± 1%	0.1%	40 mv	300 ma
(4) Regulated	+300 V dc	± 1%	1.0%	2 v	250 ma
(5) Regulated	-300 V dc	± 1%	0.1%	25 mv	20 ma

	VOLTAGE	VOLTAGE TOLERANCE	FREQUENCY	MAX VOLT-AMP
(6) Unregulated	115 V ac	± 10%	380 to 420 cps	200 va Referenced
(7) Unregulated	115 V ac	± 10%	380 to 420 cps	30 va delayed
(8) Unregulated	115 V ac	± 10%	380 to 420 cps	500 va 3Ø

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the procuring activity. When the tests of 4.2.1 are supervised, the procuring activity may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the procuring activity reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 2 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the government inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

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(2) Vibration Test (90 minutes cycling - in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the government inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	26.5 ± 2.5 volts

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

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(2) The circuit reference symbol-number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the procuring activity will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Use - The equipment shall be capable for use in naval aircraft to carry out specific military missions.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V, DC and 115 ± 1V, AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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